

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

I P. Harsha Vardhan 192311039 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

P. Harsha Vardhan
Signature of the student:

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response. (5 Marks)

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example. (5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

6. In the context of the Find-S algorithm, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only. (5 Marks)

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

7. In the context of the Find-S algorithm, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks)

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

8. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the **specific (S)** and **general (G)** boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

9. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the **version space (S and G boundaries)** and show the updates after each example. (5 Marks)

Dataset:

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. (5 Marks)

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing,

feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

12. Design a machine learning model for spam email classification using labeled data. (5 Marks)

- (a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
- (b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.
- (c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks)

Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept *EnjoySport*.

- (a) Explain the initialization of S (specific boundary) and G (general boundary).
- (b) Show how S and G change with each training example.
- (c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)

- (a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).
- (b) Explain feature scaling and handling missing data.
- (c) Describe how overfitting can be detected and avoided.

15. Explain the concept of clustering in unsupervised learning. (5 Marks)

- (a) Describe the working of the K-Means algorithm with a step-by-step example.
- (b) Explain how the value of K is chosen using the Elbow Method.
- (c) Compare K-Means with Hierarchical Clustering.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

With the necessary sketch, explain all the possible blocks in the machine-learning system. Given examples for showing the concept of supervised and unsupervised learning methods in the machine learning scheme.

Data collections

Gather raw data from Sensors DB, logs, web/APIs and surveys. Data quality and quantity set the ceiling of model performance

Data preprocessing

Clean noise and duplicates
impute missing values
encode categorical normalise
Scale and fix class imbalance

Feature Engineering

Extract create & select the most information attribute (correlations PCA importance) to reduce dimensions

Data Splitting

Divide data into training 70%
Validate ~15% & test ~15% sets
to avoid data leakage

Models Selection

Choose the algorithm for task (regressions, classifications, clustering) & its hyper parameters

Training (Learning)

Optimise the model parameters by minimising a loss functions (gradient descent) on training data.

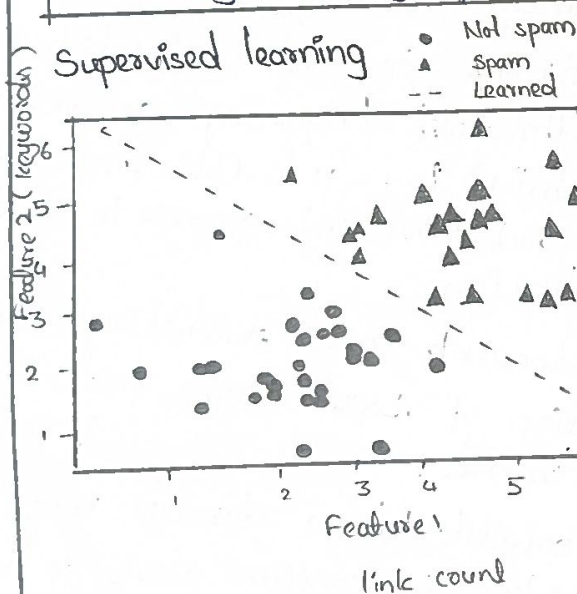
Evaluation & Tuning

Test on unseen data: accuracy, precision, recall, F1, RMSE
Tune & avoid over/under-fitting

Deployment & prediction

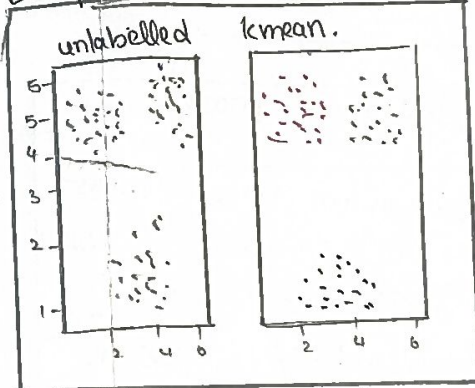
Release the model as an API/app so it predict on new real world data

Supervised learning



Idea: learn a mapping input from the labelled examples then predict labels for new data
Example: Spam detection
Emails labelled spam/Not spam
train a classifier (Naive Bayes, SVM) A new email is assigned a label.
Type: classifications, and regressions.
Algorithms: linear/logistic regression, kNN, decision tree, SVM

Unsupervised learning



Idea: No target labels; the model discovers hidden structure on its own.

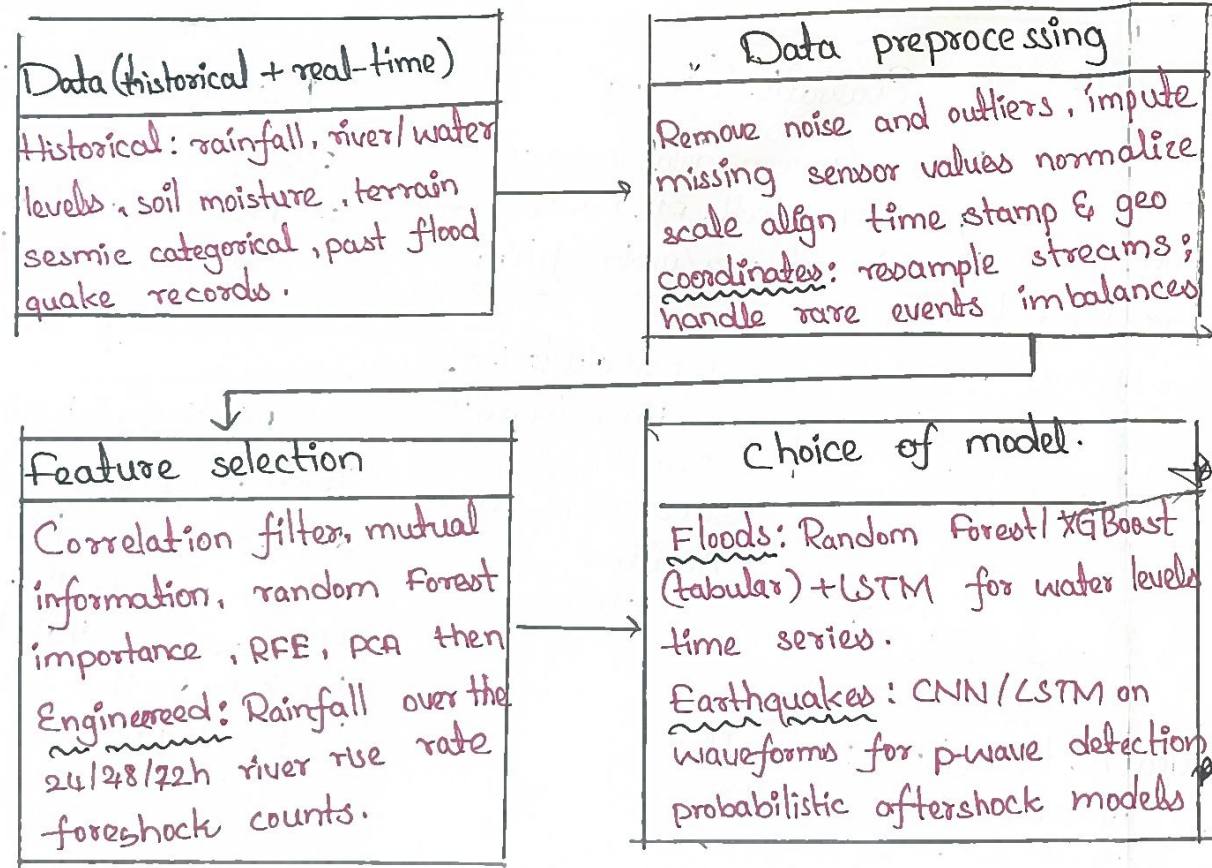
Example: Customer kmean groups shoppers by annual spend & visualize frequency into clusters with no predefined classes.

Types: clustering, dimensionality reduction (PCA), association rules
Algorithm: k means

Monitoring & Feedback

Track performance and data drift, collect new data and retrain periodically. The Loop returns that to the Block 1 so the system keeps improving.

2. Design a machine learning system for disaster management to predict events such as flood and earthquakes using historical and real-time data. Explain the choice of model, data, preprocessing and feature selection techniques.
- a) Describe the steps involved in training, validating & evaluating the model.
- b) Explain how system can assist in early warning & emergency system



2(a) Training, validation and evaluation

1. Split chronologically: Train 70% / Validate 15% / Train 15%. no shuffling so future data never leaks into trainings.
2. Train on historical data (flood and quake events plus normal periods) using class weight because disasters are rare.
3. Validation tune hyperparameters with time-series cross validation apply early stopping compare models.
4. Evaluation on the unseen test set: precision, F1, ROC-AUC / PR-AUC, false alarm rate and warning lead time.
5. Back test on past real events then develop & retain as new data arrives.

2(b) Early warning and emergency response

- Early warning: Live sensor streams feed the model, which outputs a risk score, Crossing a threshold triggering graded alerts (watch/warning/evaluation) by SMS, calls, mobiles, radios, giving hours of flood lead time and seconds to minutes for the quakes CIP wave detection.
- Emergency response: Show vulnerable zones, evacuation routes, shelter placement and allocation of rescue teams, boats, medical supplies, shelter placements.
- Feedback: post-event data & satellite/drone damage maps are used to retrain the model; humans confirm alerts to limit false alarms.

3. In the context of find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity and wind. Apply find-S algo to determine the maximally specific hypothesis show the step by step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis.

Dataset:

Example	Outlook	Temperature	Humidity	wind	Tennis
1	Sunny	Hot	High	weak	No
2	Sunny	Hot	High	strong	No
3	Overcast	Hot	High	weak	yes
4	Rainy	mild	High/Weak	weak	yes
5	Rainy	cool	Normal	weak	yes
6	Rainy	cool	Normal	strong	No
7	Overcast	mild	Normal	strong	yes
8	Sunny	mild	High	weak	No

Context & Concept:

Classify whether a day is suitable for tennis from 4 attributes

find-S: Start with the most specific hypothesis $\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$ use only positive examples. If an attributes value matches, keep it; if it differs generalise it to ?. Negative examples (1, 2, 6, 8) are ignored positive examples $\Rightarrow 3, 4, 5, 7$.

Our initial assumption

$$h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$$

from positive examples 3, 4, 5, 7

Ex 3: Overcast Hot High Weak first positive

$$h_1 = \langle \text{Overcast}, \text{Hot}, \text{High}, \text{Weak} \rangle$$

Ex 4: Rainy, Mild, High, Weak, Outlook

$$h_2 = \langle ?, ?, \text{High}, \text{weak} \rangle$$

Ex 5: Rainy, Cool, Normal, weak,

$$h_3 = \langle ?, ?, ?, \text{weak} \rangle$$

Ex 7: Overcast, Mild, Normal, Strong, Wind

$$h_4 = \langle ?, ?, ?, ? \rangle$$

find hypothesis: $h = \langle ?, ?, ?, ? \rangle$

Every attribute has been generalised. so, it accepts any day. The data are not consistent with a single conjunctive concept and this h also matches the negative examples. which find S never checks.

4. In the context of the candidate elimination algorithm consider a dataset used to classify whether a person will enjoy a sport based on attributes such as whether Temp, Humidity and wind. Using the candidate elimination algorithm, determine the version space by computing both the specific and general boundaries. Show the step by step updates after processing each training examples.

Example	Weather	Temperature	Humidity	Wind	Sport
1	Sunny	Warm	Normal	weak	Yes
2	Sunny	Warm	High	strong	Yes
3	Rainy	Cold	High	weak	No
4	Sunny	Cold	Normal	weak	Yes
5	Rainy	Warm	Normal	strong	No

Context and Concept:-

Problem: Find every hypothesis consistent with the data (the version space) bounded by S (most specific) and G (most general)

start: $S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$ $G_0 = \langle ?, ?, ?, ? \rangle$

Positive examples: Generalise S minimally to cover it: delete members of G that do not cover it.

Negative Examples: keep S if it excludes the examples specialise each G member minimally so it excludes the example while staying more general than S .

Specialising G : can only use values taken from S : an attributes where S has ? cannot be used.

Converged when $S = G$

Ex 1: $\langle \text{Sunny, Warm, Normal, Weak} \rangle$ Copy $\neq$ in S (positive)

$S_1 = \langle \text{Sunny, Warm, Normal, weak} \rangle$

$G_1 = \langle ?, ?, ?, ? \rangle$

Ex 2: $\langle \text{sunny, warm, High, Strong} \rangle$ Humidity & Wind differ (positive)

$G_2 = \langle ?, ?, ?, ? \rangle$

$S_2 = \langle \text{Sunny, Warm, ?, ?} \rangle$

Ep 3: $\langle \text{Rainy, cold, High, weak} \rangle$

S_2 already excludes it so $S_3 = S_2$. Specialise G using S_3 (Negative)
values only (Humidity and wind are ? in S , so no options there)

$G_3 = \langle \text{Sunny, ?, ?, ?} \rangle$

$\langle ?, \text{Warm, ?, ?} \rangle$

Ep 4: $\langle \text{sunny, cold, Normal, weak} \rangle$ Temp: warm vs cold (positive)

$S_4 = \langle \text{Sunny, ?, ?, ?} \rangle$ Remove: $\langle ?, \text{warm, ?, ?} \rangle$

$G_4 = \langle \text{Sunny, ?, ?, ?} \rangle$

Ep 5: $\langle \text{Rainy, warm, Normal, strong} \rangle$

S_4 & G_4 already exclude it (Rainy $\neq$ Sunny).

No change

$S_5 = G_5 = \langle \text{sunny, ?, ?, ?} \rangle$

(Negative)

5. In the context of the find-S Algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, student and Credit Rating. Apply the find-S Algorithm to determine the maximally specific hypothesis show step by step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.

Dataset:

Example	Age	Income	Student	Credit Rating	Buy
1	young	High	No	Fair	No
2	young	High	No	Excellent	No
3	young Middle	High	No	Fair	Yes
4	young Senior	Medium	No	Fair	Yes

1. Context and Concept:

Problem context: Find the maximally specific hypothesis for buyers using only positive examples (3 and 4) negatives (1, 2) are ignored by find-S.

Concept: Start from $\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$ keep matching values replace mismatches with ?.

Check: Ex 1 (Young, High, No, Fair) is a negative case but matches the final h. so, pure conjunction cannot separate the data perfectly.

Initial values:

$$h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$$

As per find-S, we are ignoring negative rows both 1 and 2 are removed.

Ex 3 (positive)

$\langle \text{Middle, High, No, Fair} \rangle$ first positive so copy it.

$$h_1 = \langle \text{Middle, High, No, Fair} \rangle$$

Ex 4 (positive)

$\langle \text{Senior, Medium, No, Fair} \rangle$ Age & Income differ $\rightarrow$?

$$h_2 = \langle ?, ?, \text{No, Fair} \rangle$$

Final hypothesis:

$h = \langle ?, ?, \text{No, Fair} \rangle$ A customer buys when they are not a student & have fair credit.

Negative (Ex 1, 2) were ignored.